Julius Alexander Zeiss, M.Sc. Physics

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Profile

- Research in quantum information theory at its interface with convex optimization and mathematical physics: quantum de Finetti theorems, symmetry-reduced semidefinite programming hierarchies, and efficient algorithms with certifiable guarantees for quantum correlations, approximate quantum error correction, and polynomial optimization
- Speaker at leading international conferences (TQC, Beyond IID, AQIS) and invited speaker at institutions including Stanford University and the Perimeter Institute for Theoretical Physics
- Host of a regular A.I. Workshop at RWTH Aachen on AI tools for rigorous, reproducible science

Education

since 03/2023	**PhD in Physics (Quantum Information Theory), RWTH Aachen University, Germany** Institute for Quantum Information, final-year PhD student Advisor: Prof. Dr. Mario Berta
10/2019 – 08/2022	**M.Sc. Physics, University of Cologne & University of Bonn, Germany** Focus: Theoretical Particle Physics, General Relativity, Mathematical Physics Thesis: *SDP Relaxations for Quantum Inflation* Advisor: Prof. Dr. David Gross, Thesis Grade: 1.0 (very good) Overall Grade: 1.4 (very good)
10/2018 – 08/2019	**M.Sc. Physics, University of Münster, Germany** Focus: Theoretical Nuclear & Particle Physics, Theoretical Non-linear Physics, Quantum Information Theory, Biophysics
10/2015 – 09/2018	**B.Sc. Physics, University of Münster, Germany** Thesis: *Numerical Treatment of the Manning Potential for the Description of Ammonia* Advisor: Prof. Dr. Gernot Münster, Thesis Grade: 1.0 (very good) Overall Grade: 1.8 (good)
04/2013 – 09/2015	**B.Sc. Business Administration, University of Münster, Germany** (discontinued, transferred to Physics)
09/2010 – 05/2012	**International Baccalaureate, King William's College, Isle of Man** Overall Grade: 41/45 Points (very good, 1.1 German Abitur)

Publications and Preprints

2026	G. Koßmann, J. A. Zeiss, *Fixed points in de Finetti hierarchies*, arXiv:2607.23689
2026	J. A. Zeiss, G. Koßmann, R. Schwonnek, M. Plávala, *Finite de Finetti for convex bodies and polynomial optimization*, arXiv:2601.15184
2025	J. A. Zeiss, G. Koßmann, O. Fawzi, M. Berta, *Approximating fixed size quantum correlations in polynomial time*, arXiv:2507.12302
2025	G. Koßmann, J. A. Zeiss, O. Fawzi, M. Berta, *On approximate quantum error correction for symmetric noise*, arXiv:2507.12326

Invited Talks

06/2026 **Stanford University, CS Theory Group, Palo Alto, USA**
Information-theoretic de Finetti theorems

04/2026 **University of Cologne, Quantum Information Seminar, Germany**
Approximating fixed size quantum correlations in polynomial time

07/2023 **Siegen University, Germany**
Inner approximation schemes for approximate quantum error correction

02/2023 **Perimeter Institute for Theoretical Physics, Waterloo, Canada**
Infinite Dimensional Optimisation Problems in Quantum Information — An operator algebra approach to the NPA Hierarchy

05/2022 **Alpen-Adria-Universität Klagenfurt, Austria**
SDP methods for the quantum causal compatibility problem

Contributed Talks

09/2026 **TQC 2026, Sherbrooke, Canada**
A Family of Information-Theoretic de Finetti Theorems for Constrained Optimization

05/2026 **ILAS 2026, Virginia Tech, Blacksburg, USA**
Convergent Inner-Outer Approximation Schemes From De Finetti Theorems For Games And Quantum Error Correction

03/2026 **Quantum at the Dunes, IIP-UFRN, Natal, Brazil**
Approximating fixed size quantum correlations in polynomial time

08/2025 **AQIS 2025, The University of Hong Kong, Hong Kong**
Approximating fixed size quantum correlations in polynomial time

07/2025 **13th Beyond IID in Information Theory, TU Munich, Germany**
Approximating fixed size quantum correlations in polynomial time

Posters

04/2026 **851. WE-Heraeus-Seminar: 90 Years of Einstein-Podolsky-Rosen Paradox, Bad Honnef, Germany**
Approximating fixed size quantum correlations in polynomial time

01/2026 **QIP 2026, Riga, Latvia**
Approximating fixed size quantum correlations in polynomial time

08/2025 **QMATH Masterclass on Representation Theory in Quantum Information Science, University of Copenhagen, Denmark**
Schur–Weyl Duality for Efficient Approximation of Constrained Separability

07/2024 **Beyond IID in Information Theory, University of Illinois Urbana-Champaign, USA**
Approximating fixed size quantum correlations in polynomial time

Teaching

since 03/2023 **RWTH Aachen University, Germany**
Teaching and supervision of students at the Institute for Quantum Information

04/2021 – 04/2022 **University of Cologne, Germany**
Tutorials on Linear Algebra & Analysis, Proofs in Mathematics, Theoretical Physics

04/2019 – 09/2019 **University of Münster, Germany**
Tutorials on Applied Physics

Academic Service

Reviewing TQC (2023, 2024, 2025) • ISIT (2023, 2024, 2025) • QIP (2024) • Beyond IID (2023, 2024)

Outreach

since 2025	**A.I. Workshop, RWTH Aachen University, Germany** Founder and host of a regular workshop on AI tools for research — from coding assistants and version control to formally verified proofs — and on using them to conduct better science
2024 – 2026	**ML4Q — Matter and Light for Quantum Computing, Germany** Instructor: school internships (July 2024), Quanteen Day 2024, Quanteen Day 2026
11/2022	**DPG Physik Journal** A. Matthies and J. Zeiss, *Quantum Computing*, review article on the Quantum Computing Bad Honnef Physics School, Physik Journal 21 (2022) No. 11, p. 57

Skills

Languages	German (native), English (bilingual)
Coding	Python, Mathematica, Fortran, PHP, JavaScript, HTML, SQL, VBA

Outside of Academia — Employment

2014, 2015	**Fielmann AG, Hamburg, Germany** — DCF Modelling with Monte-Carlo Simulations, VBA & SQL Solutions
01/2013 – 02/2013	**HMT, Hamburg, Germany** — Mathematical Modelling using Monte-Carlo Simulations
08/2011	**AKO Capital, London, England** — Internship in Financial Research using Mathematical Modelling